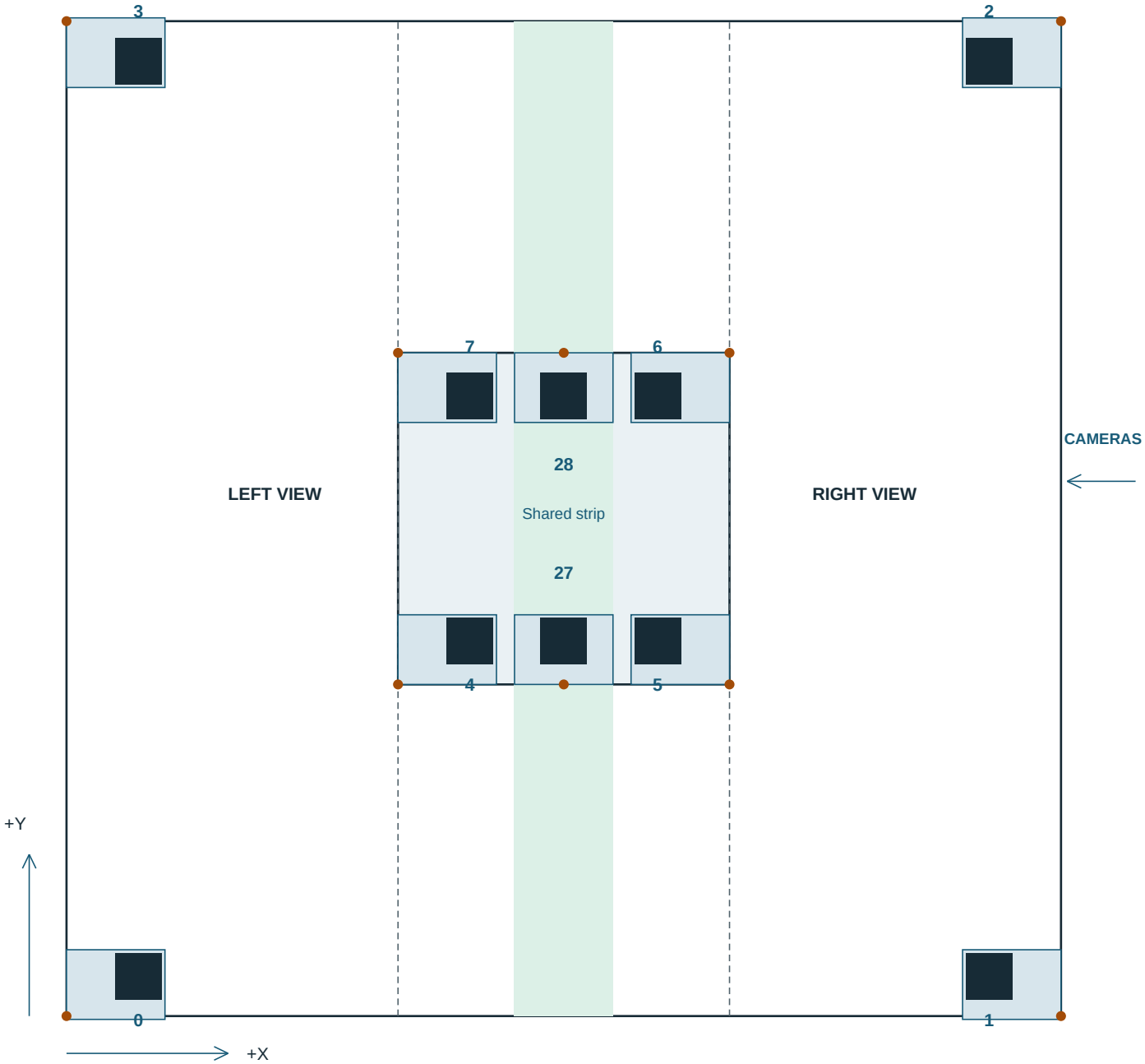
Side-view tracking targets

PRINT-READY A4 | v3 | shared sheets aligned at inner-square edges



Orange = R reference; sheet outlines are schematic, not print-scale.

Print first

Pages **4-11**: eight fixed references. Page **12**: movable ID 8. A4, single-sided, Actual size / 100%, no Fit or borderless enlargement.

Then calibrate

Page **15**: ChArUco lens board. Pages **13-14**: optional shared references. Page **16**: optional AprilTag comparison, not a default target. Page **17**: setup record. Only pages 13-14 use landscape A4; all others are portrait.

Reference meaning

Origin = proposed near-left inside-wall corner; +X right, +Y far. All listed field positions are placement targets, not a survey. Keep earlier packs for their existing recordings.

Align the references

The sheet guides define placement; they do not certify a perfect field.

1 Set the physical frame

Use a consistent straight centre datum. The measured outer spans of 2994 mm suggest 3 mm outward at each end **if their shortening is centred**. Desired inside-wall faces are 1500 physical mm either side. Barrier lengths alone do not establish inside clearance or squareness. The accepted survey allowance is approximately 1-2 mm.

2 Place the outer sheets

Turn each page as labelled. Its **short paper edge** aligns with the left/right blue/white reference boundary: ID 0 at left, ID 1 at right (IDs 3 and 2 likewise at the far end). Use the short edge shown at the PDF bottom, not the page top. The **solid REF** line represents the near/far inside wall face. The **dashed MAT** line is 3 mm towards the field interior and aligns with the printed blue/white edge. Thus the paper extends 13 mm into the printed blue area, but the proposed wall correction is only 3 mm.

3 Identify R correctly

On outer sheets R is the intersection of the solid REF line and the physical short edge, **not the paper corner**. On inner sheets R is the labelled physical paper corner. All long page directions point across X into their allowed area; codes are never mirrored.

4 Place the inner references

Keep the complete sheets inside the central non-driving area. Use the physical inner-wall/reference geometry where possible. Printed inner corners still carry longitudinal scale uncertainty. Record real R coordinates if they differ from the nominal targets; do not hide a coordinate error by bending the camera transform. **Shared pages 13-14:** use landscape A4. Their physical bottom long edge touches the inner near/far blue/white boundary; keep the paper inside the central square. Centre the arrow guide on the bottom-to-top field line (nominal X=1500 mm). R is the physical long-edge midpoint. ID 27 is unturned at the near edge; turn ID 28 by 180 degrees at the far edge. The arrow stops 6 mm short of the edge to remain printable.

5 Keep the capture geometry stable

Test all code corners with the barriers installed. Fix pan, tilt, optical zoom, focus and capture mode during recording. Calibrate each camera's lens distortion separately from registering it into this shared field frame. Correct a raised robot marker for its height and axle offset.

6 Validate independently

Use ID 8 or a known rigid displacement that was not used to fit the references. Flat matte backing, short exposure and native camera video matter. A low fitting residual, a quiet marker or agreement between two fitted views is not proof of below-5-mm absolute accuracy. Moving hand-held paper tests blur, not independent dynamic truth.

Marker precision choices

Research reviewed 03 October 2026 | recommendations, not measured performance

Large ArUco for the working tracker

Keep the existing 4x4/50 family and 140 mm fixed codes. More code bits can help identification but require more resolved detail [1]. Our engineering choice is large simple tags, clear white margins and qualified corner refinement, rather than replacing everything.

ChArUco for lens calibration and precision trials

Checkerboard intersections can be refined more accurately than isolated tag corners [2]. The 5x7 board has 36 mm squares and 24 mm tags. Bring it close enough to resolve the smaller tags; full-half-field visibility is not assumed. This board needs its own detector.

Larger movable target and an honest comparison

Default ID 8 is 180 mm, 29% larger in side length than the old 140 mm test code. This is a sampling advantage, not a promised 29% accuracy improvement. Compare the optional 180 mm AprilTag 36h11 at identical position, exposure and view. AprilTag's authors recommend Standard41h12; 36h11 here is selected for installed OpenCV compatibility [1,3].

Other systems remain possible

STag uses an inner circle to refine pose stability [4]. CCTag uses coded concentric rings for centre localization under challenging blur [5]. They need different detectors and, for circular centres, additional geometry for heading/pose. Revisit if blur or corner jitter remains limiting.

Where the error budget actually goes

At 3.2 m / 1920 pixels, ideal sampling is 1.67 mm/pixel: 140 mm spans 84 pixels, 180 mm spans 108. Perspective changes this by region. Subpixel detection cannot remove mat/reference bias, lens distortion, flex, target height or timestamp error. At 2 m/s, 1 ms is 2 mm.

Primary sources

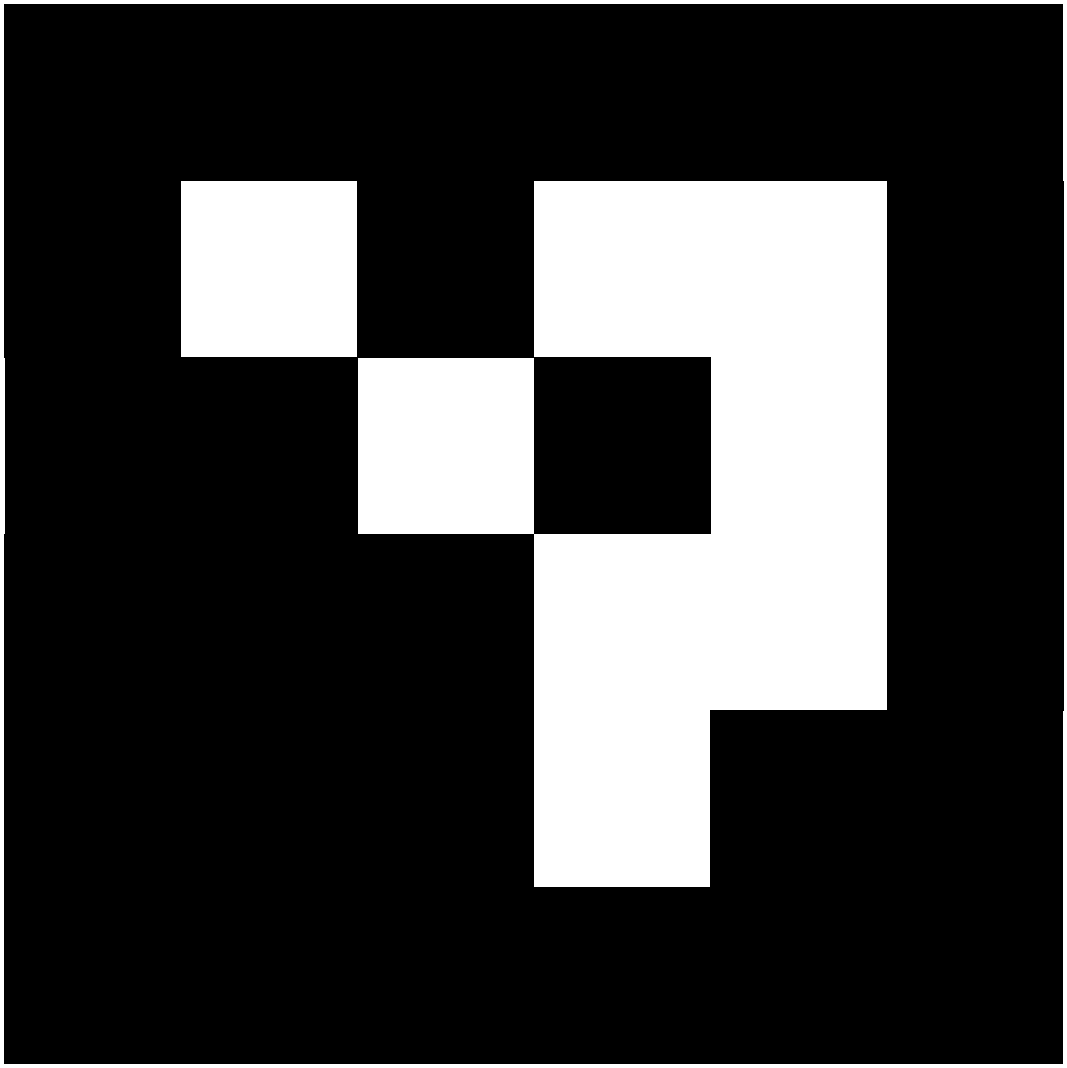
[1] OpenCV ArUco detection - click to open

[2] OpenCV ChArUco detection - click to open

[3] AprilTag author repository - click to open

[4] STag research paper - click to open

[5] CCTag author documentation - click to open



SOLID = REF WALL / DASHED = MAT EDGE

ID 0 | OUTER NEAR LEFT

Turn whole sheet 270 deg CCW from this page view.

Long edge points RIGHT (+X); short edge runs along field Y.

Short paper edge: LEFT boundary. Solid REF: NEAR inside wall.

Dashed MAT: printed blue/white edge, 3 mm inside REF.

R = solid REF / short paper edge intersection. Do not trim.

R target: (0, 0) mm; NOT surveyed.

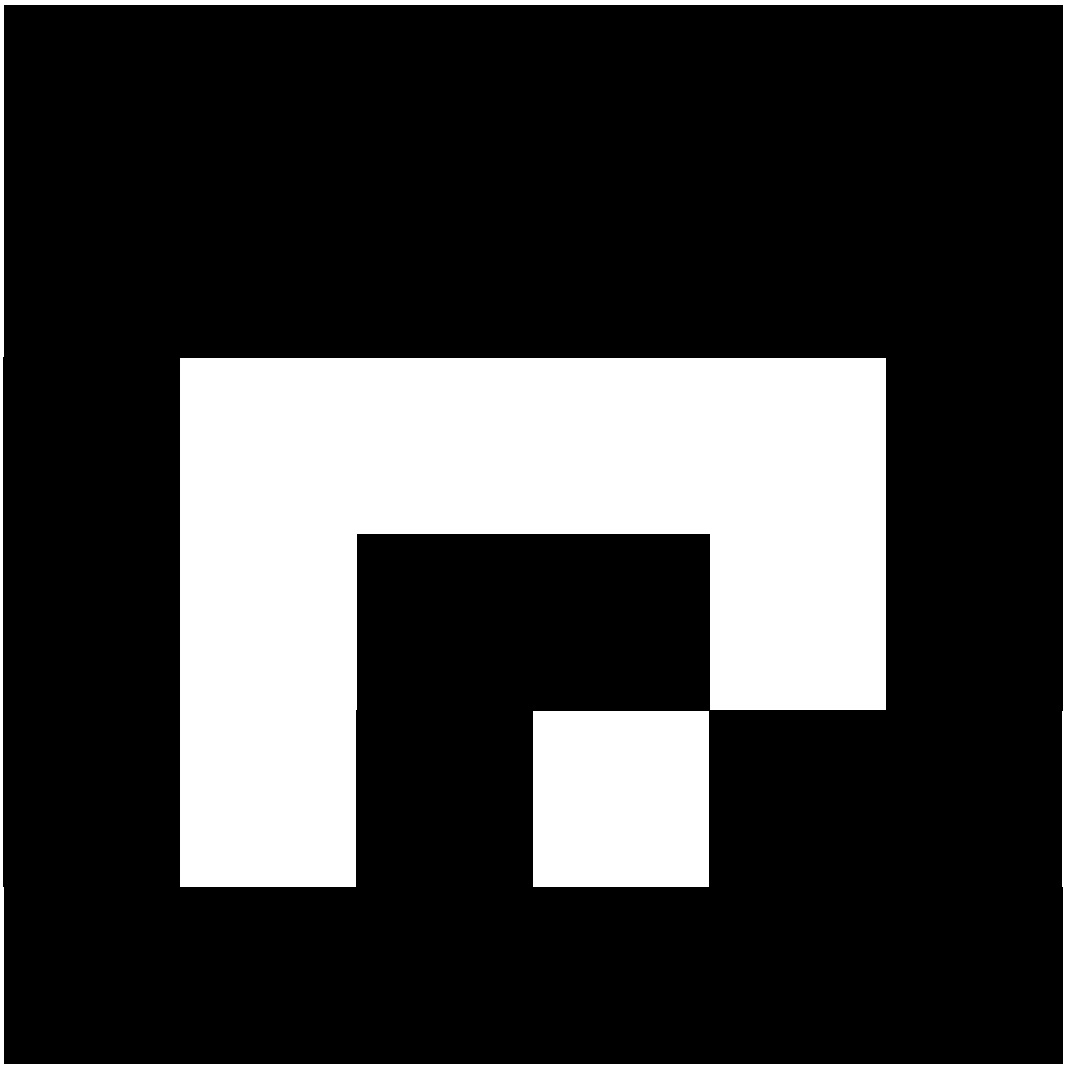
Code centre target: (217, 120) mm.

140 mm black square | DICT_4X4_50 | 100% A4

100 mm horizontal print check

R defines the coordinate anchor; keep the page flat and fixed.

SOLID = REF WALL / DASHED = MAT EDGE



ID 1 | OUTER NEAR RIGHT

Turn whole sheet 90 deg CCW from this page view.

Long edge points LEFT (-X); short edge runs along field Y.

Short paper edge: RIGHT boundary. Solid REF: NEAR inside wall.

Dashed MAT: printed blue/white edge, 3 mm inside REF.

R = solid REF / short paper edge intersection. Do not trim.

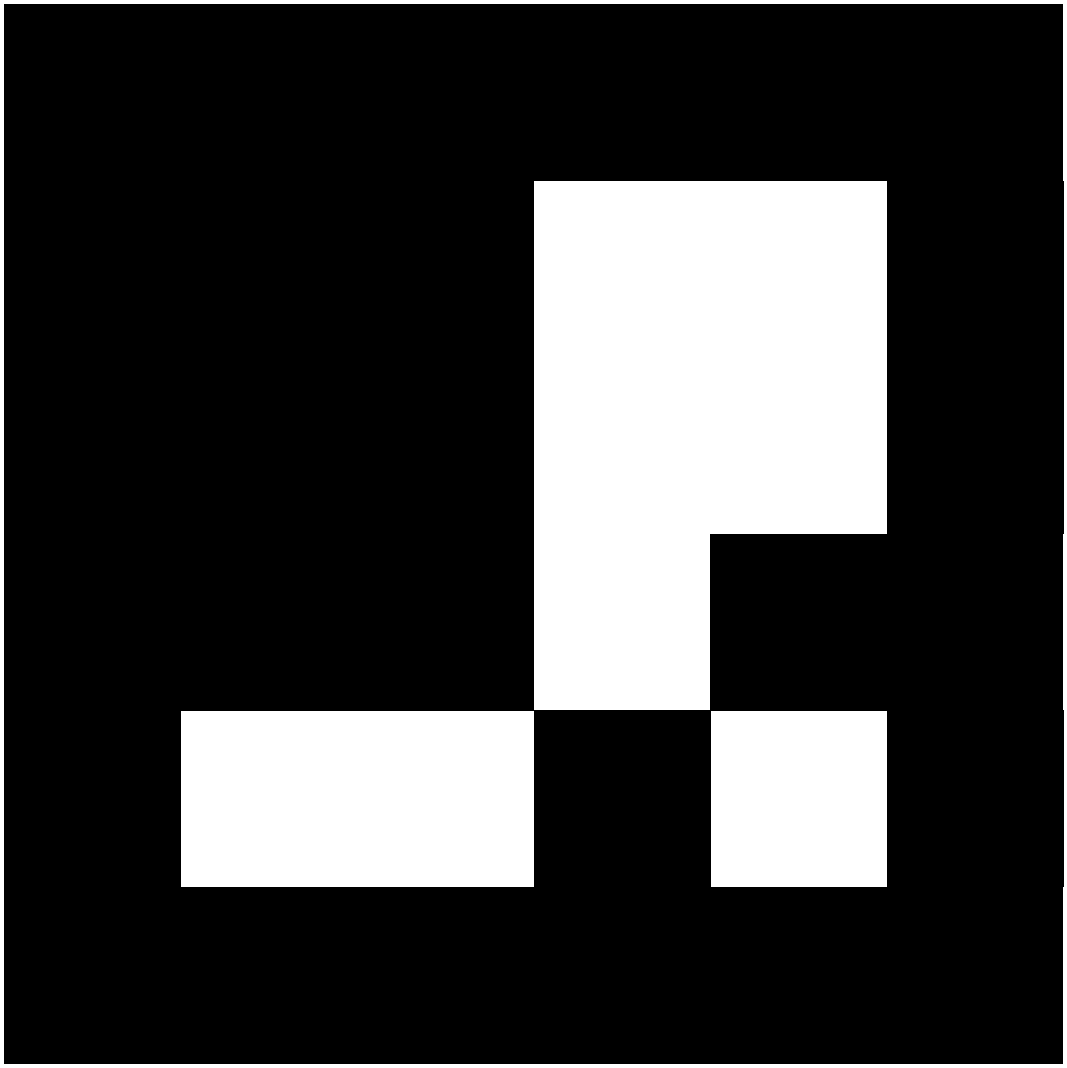
R target: (3000, 0) mm; NOT surveyed.

Code centre target: (2783, 120) mm.

140 mm black square | DICT_4X4_50 | 100% A4

100 mm horizontal print check

R defines the coordinate anchor; keep the page flat and fixed.



SOLID = REF WALL / DASHED = MAT EDGE

ID 2 | OUTER FAR RIGHT

Turn whole sheet 90 deg CCW from this page view.

Long edge points LEFT (-X); short edge runs along field Y.

Short paper edge: RIGHT boundary. Solid REF: FAR inside wall.

Dashed MAT: printed blue/white edge, 3 mm inside REF.

R = solid REF / short paper edge intersection. Do not trim.

R target: (3000, 3000) mm; NOT surveyed.

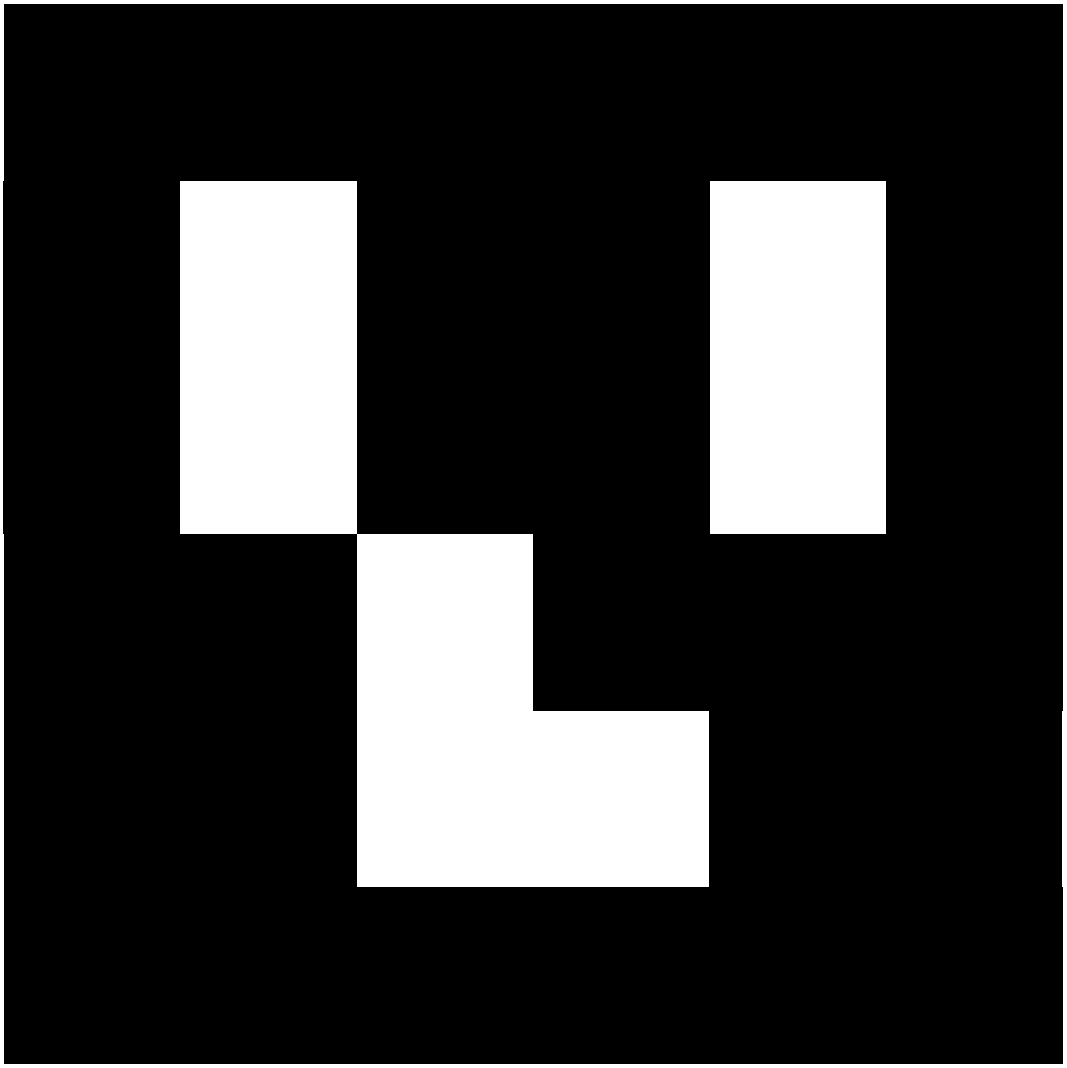
Code centre target: (2783, 2880) mm.

140 mm black square | DICT_4X4_50 | 100% A4

100 mm horizontal print check

R defines the coordinate anchor; keep the page flat and fixed.

SOLID = REF WALL / DASHED = MAT EDGE



ID 3 | OUTER FAR LEFT

Turn whole sheet 270 deg CCW from this page view.

Long edge points RIGHT (+X); short edge runs along field Y.

Short paper edge: LEFT boundary. Solid REF: FAR inside wall.

Dashed MAT: printed blue/white edge, 3 mm inside REF.

R = solid REF / short paper edge intersection. Do not trim.

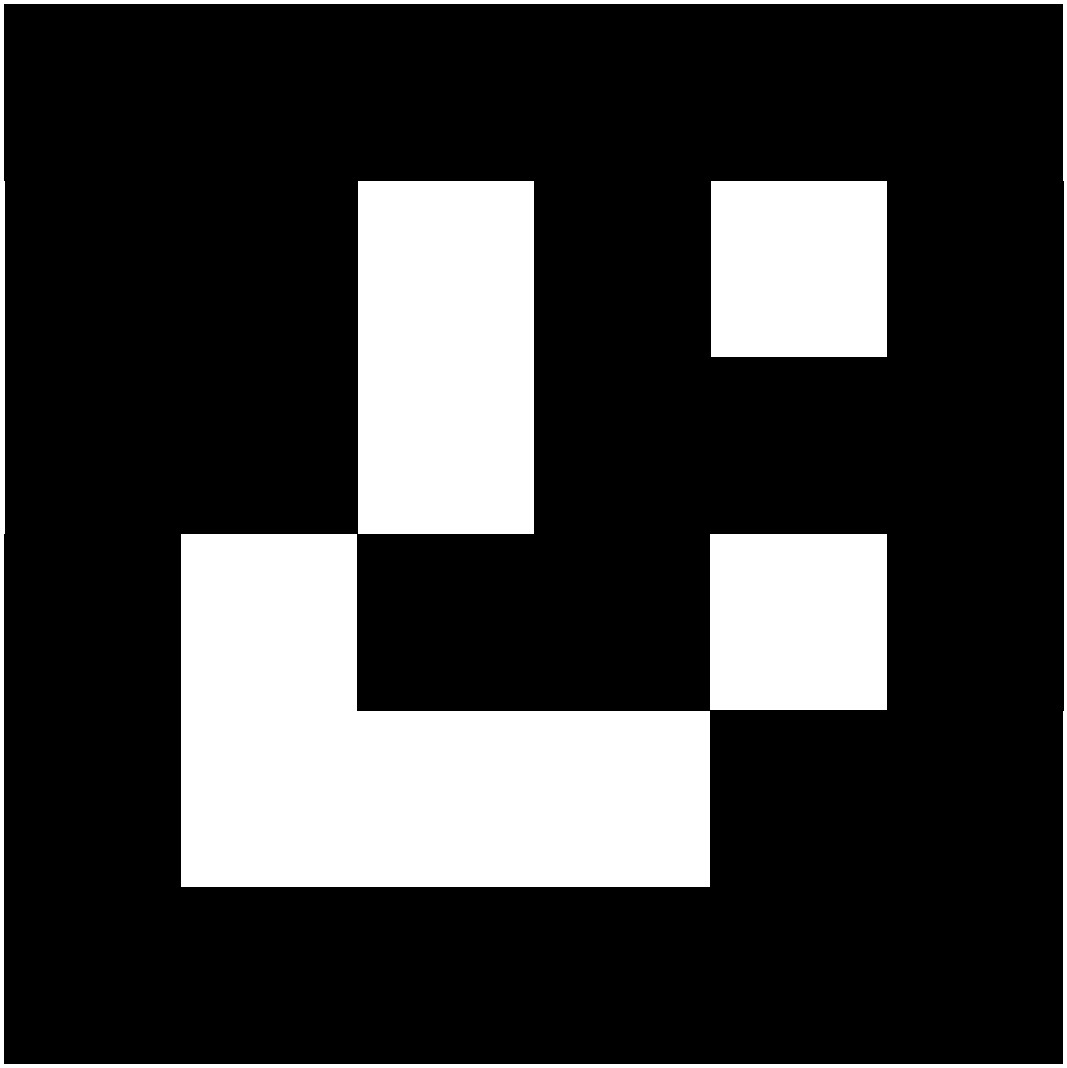
R target: (0, 3000) mm; NOT surveyed.

Code centre target: (217, 2880) mm.

140 mm black square | DICT_4X4_50 | 100% A4

100 mm horizontal print check

R defines the coordinate anchor; keep the page flat and fixed.



ID 4 | INNER NEAR LEFT

Turn whole sheet 270 deg CCW from this page view.

Long edge points RIGHT (+X); short edge runs along field Y.

Keep paper wholly inside central non-driving square.

R = RIGHT corner of the short edge shown at page bottom.

Use real inner reference geometry; printed corners are approximate.

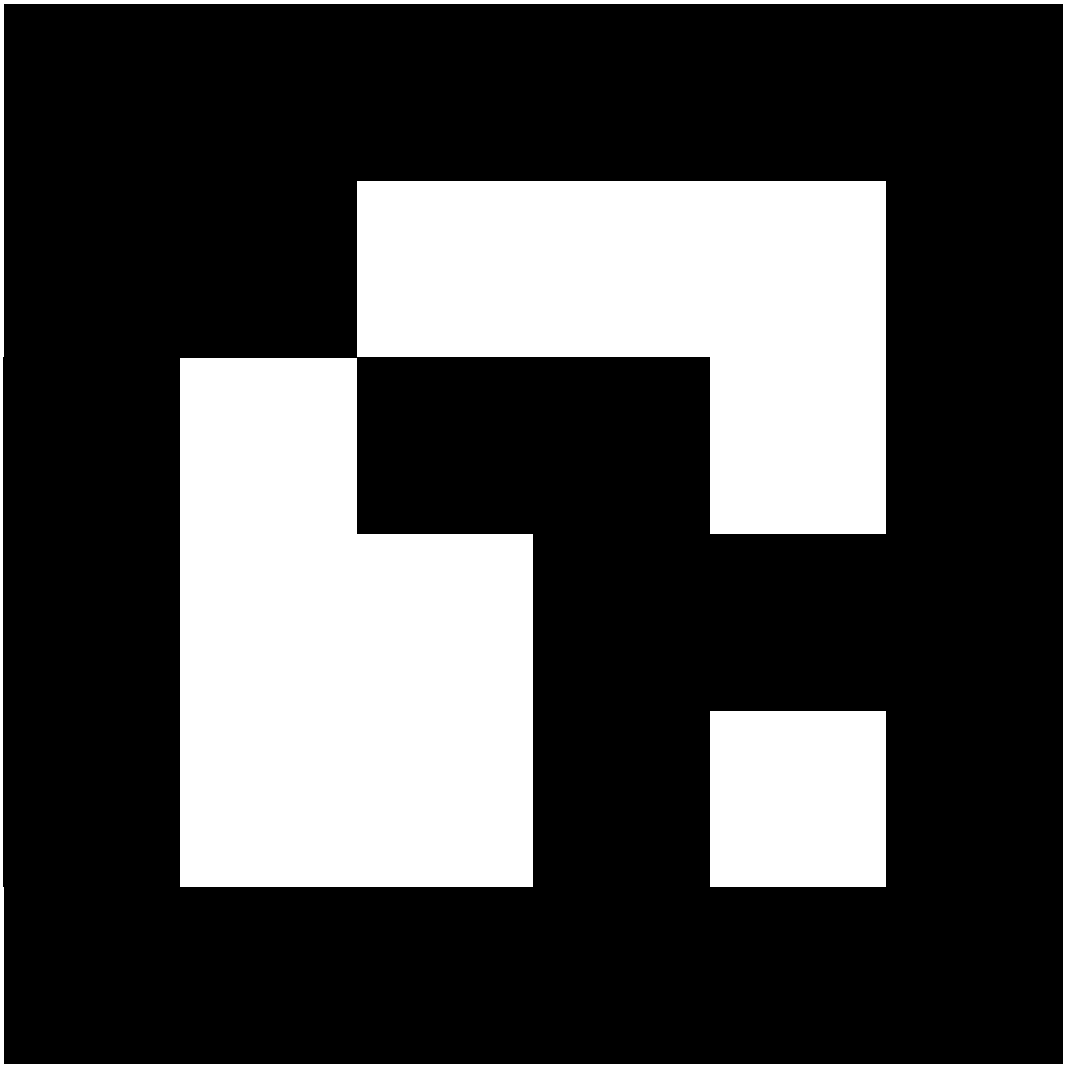
R target: (1000, 1000) mm; NOT surveyed.

Code centre target: (1217, 1130) mm.

140 mm black square | DICT_4X4_50 | 100% A4

100 mm horizontal print check

R defines the coordinate anchor; keep the page flat and fixed.



ID 5 | INNER NEAR RIGHT

Turn whole sheet 90 deg CCW from this page view.

Long edge points LEFT (-X); short edge runs along field Y.

Keep paper wholly inside central non-driving square.

R = LEFT corner of the short edge shown at page bottom.

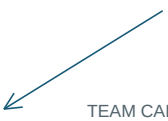
Use real inner reference geometry; printed corners are approximate.

R target: (2000, 1000) mm; NOT surveyed.

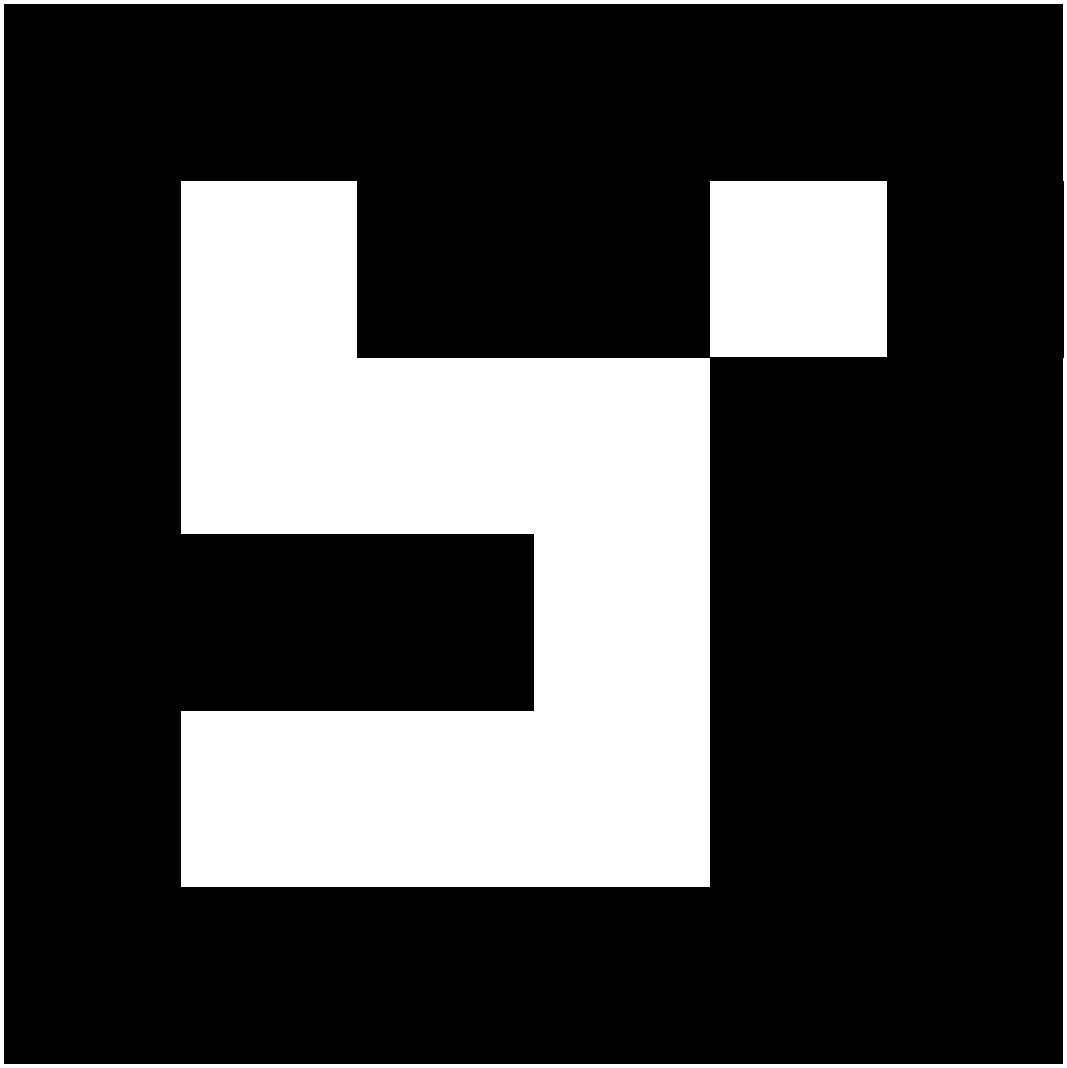
Code centre target: (1783, 1130) mm.

140 mm black square | DICT_4X4_50 | 100% A4

100 mm horizontal print check



R defines the coordinate anchor; keep the page flat and fixed.



ID 6 | INNER FAR RIGHT

Turn whole sheet 90 deg CCW from this page view.

Long edge points LEFT (-X); short edge runs along field Y.

Keep paper wholly inside central non-driving square.

R = RIGHT corner of the short edge shown at page bottom.

Use real inner reference geometry; printed corners are approximate.

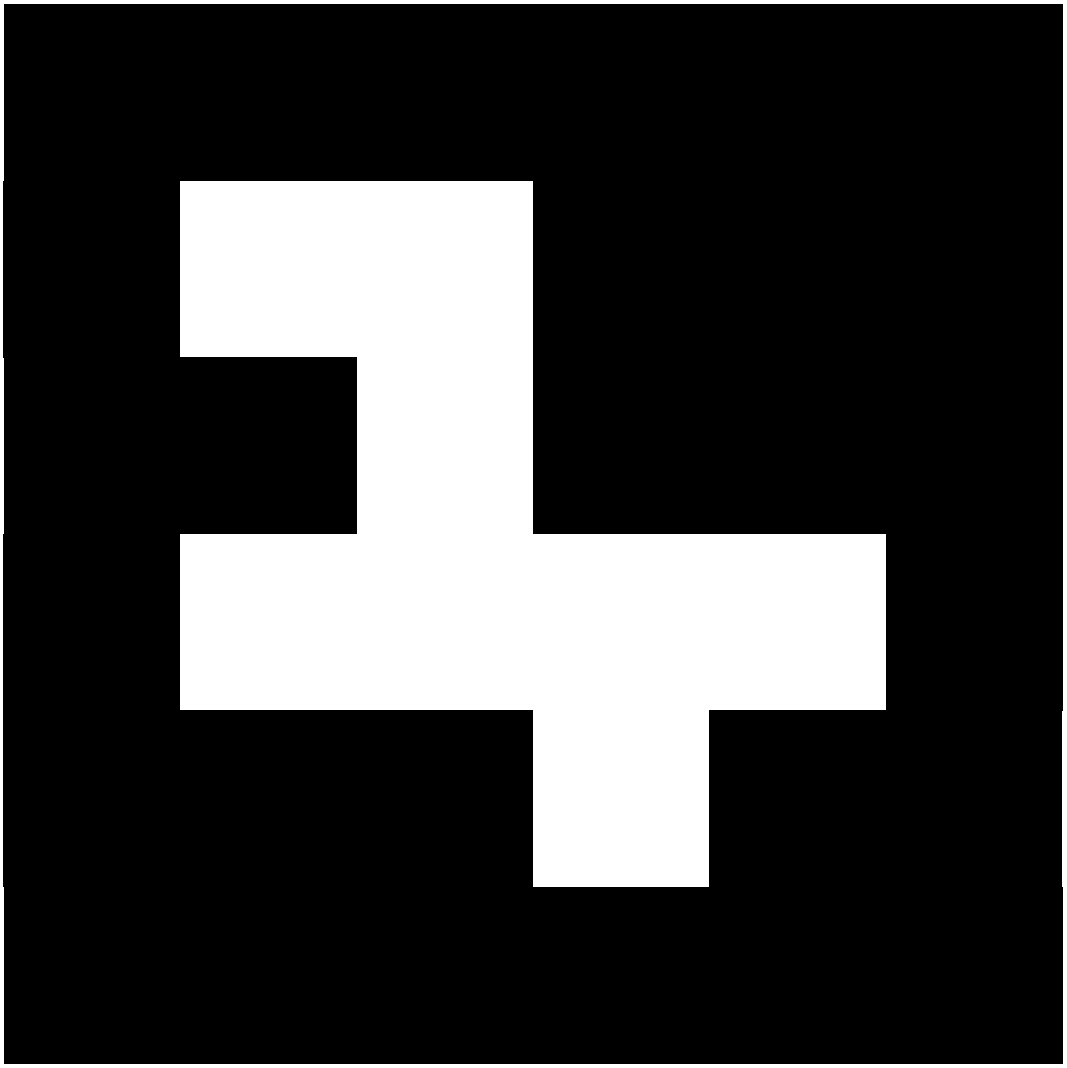
R target: (2000, 2000) mm; NOT surveyed.

Code centre target: (1783, 1870) mm.

140 mm black square | DICT_4X4_50 | 100% A4

100 mm horizontal print check

R defines the coordinate anchor; keep the page flat and fixed.



ID 7 | INNER FAR LEFT

Turn whole sheet 270 deg CCW from this page view.

Long edge points RIGHT (+X); short edge runs along field Y.

Keep paper wholly inside central non-driving square.

R = LEFT corner of the short edge shown at page bottom.

Use real inner reference geometry; printed corners are approximate.

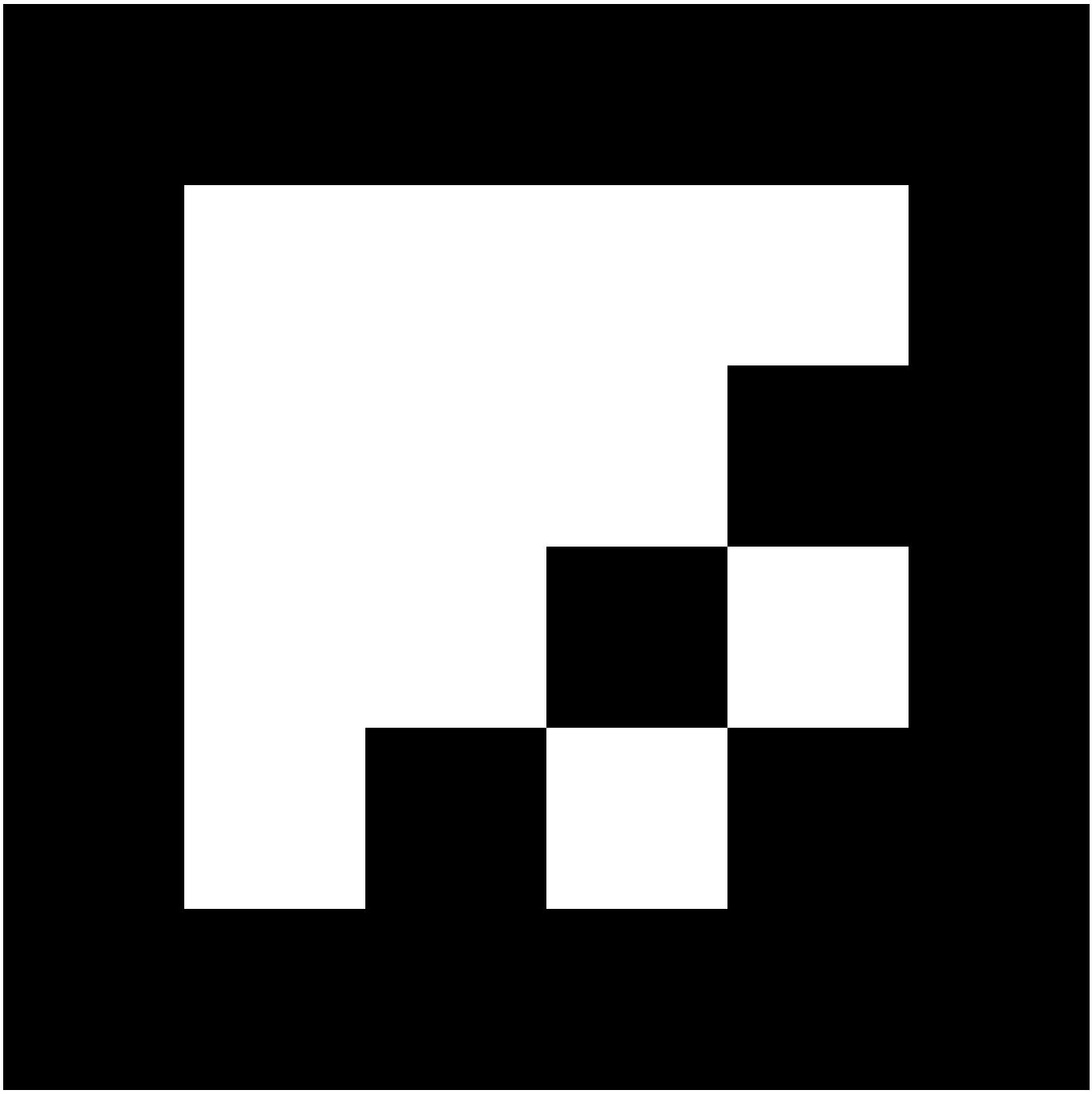
R target: (1000, 2000) mm; NOT surveyed.

Code centre target: (1217, 1870) mm.

140 mm black square | DICT_4X4_50 | 100% A4

100 mm horizontal print check

R defines the coordinate anchor; keep the page flat and fixed.



ID 8 | MOVABLE TEST TARGET

180 x 180 mm black square; keep the entire white margin clear.

R = physical bottom-left paper corner. Do not trim.

Code centre from R: (+105, +197) mm. Flat floor test only.

Held-out target: never use its test positions to refit fixed references.

Use the v3 coordinate file; old 140 mm offsets are different.

100 mm horizontal print check

R = PAPER CORNER

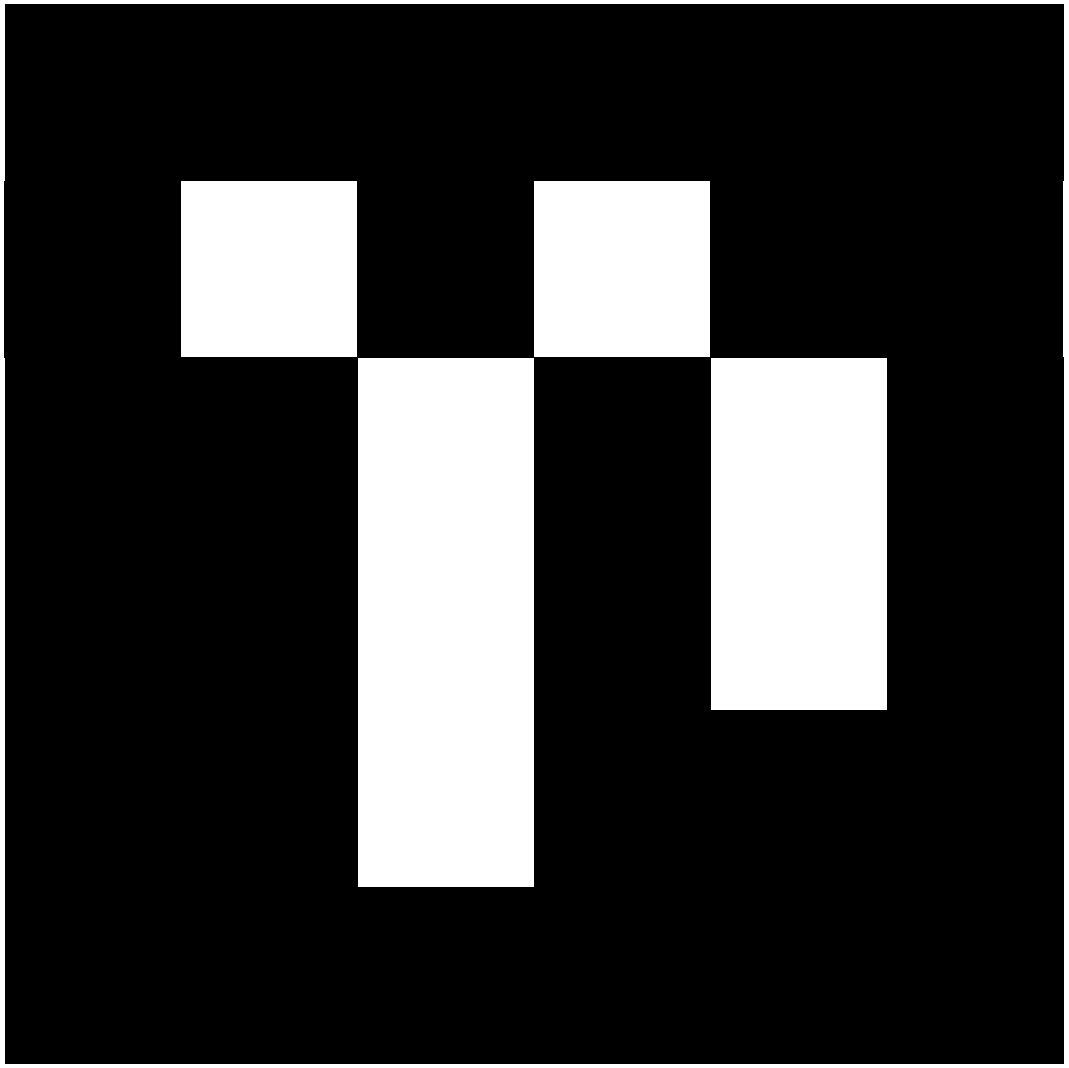
ID 27

SHARED NEAR

140 mm code
DICT_4X4_50

Landscape A4
Actual size / 100%
Do not trim

Keep the whole sheet inside the central non-driving square. Both cameras must see all four code corners.



R target: (1500, 1000) mm

Code centre: (1500, 1130) mm

R = physical long-edge MIDPOINT

BOTTOM PAPER EDGE = INNER BLUE/WHITE BOUNDARY

ALIGNMENT

No turn from this PDF view.

Put the **physical bottom long edge** on the **near** inner blue/white boundary.

Align the arrow's centre guide to the bottom-to-top field centreline.

Printed geometry remains approximate: survey R before adding this marker to the fit.

CENTRE GUIDE

148.5 mm from either short edge

Tip is 6 mm above R for print margins

Use the paper edge, NOT the arrow tip, as R

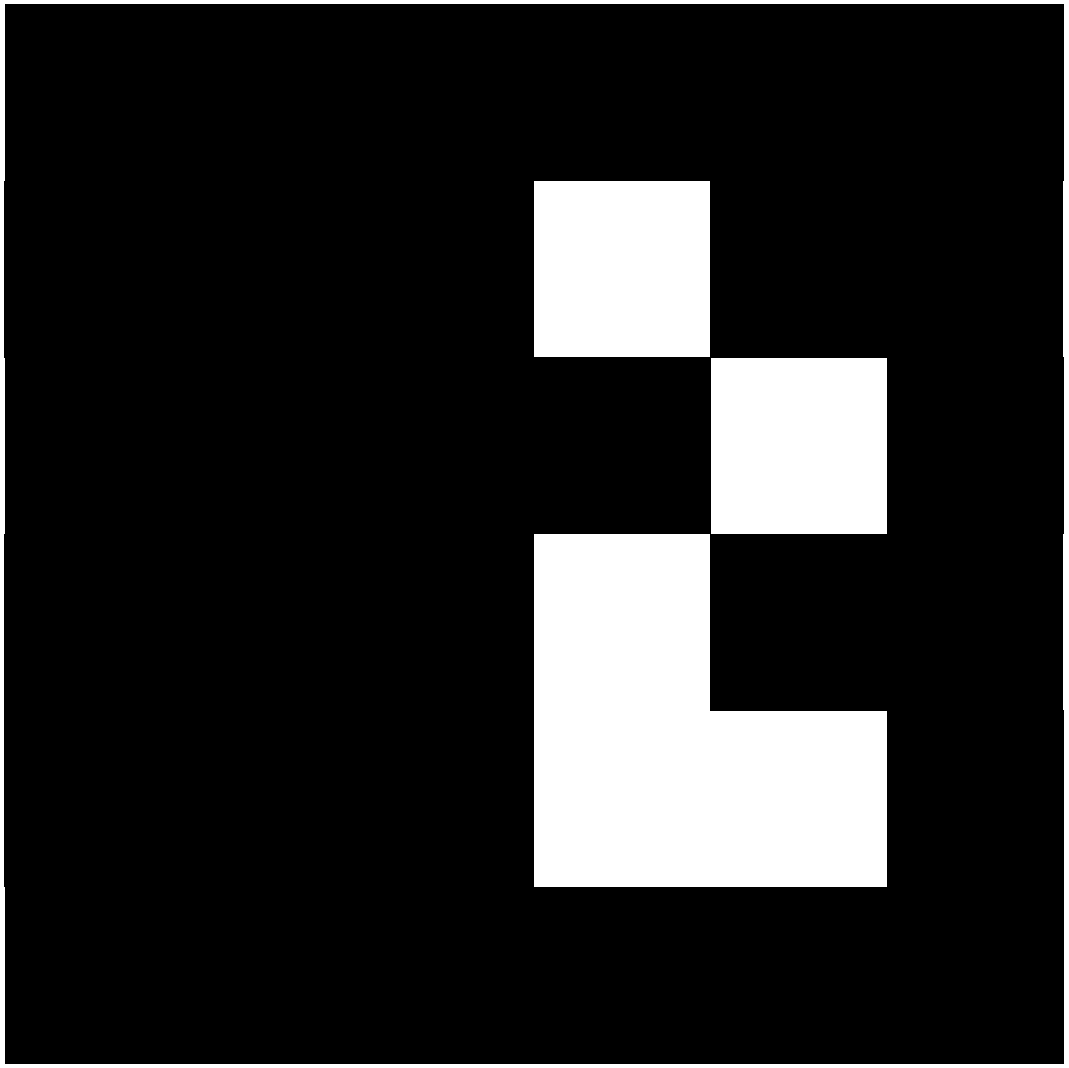
ID 28

SHARED FAR

140 mm code
DICT_4X4_50

Landscape A4
Actual size / 100%
Do not trim

Keep the whole sheet inside the central non-driving square. Both cameras must see all four code corners.



R target: (1500, 2000) mm

Code centre: (1500, 1870) mm

R = physical long-edge MIDPOINT

BOTTOM PAPER EDGE = INNER BLUE/WHITE BOUNDARY

ALIGNMENT

Turn 180 degrees from this PDF view.

Put the **physical bottom long edge** on the **far** inner blue/white boundary.

Align the arrow's centre guide to the bottom-to-top field centreline.

Printed geometry remains approximate: survey R before adding this marker to the fit.

CENTRE GUIDE

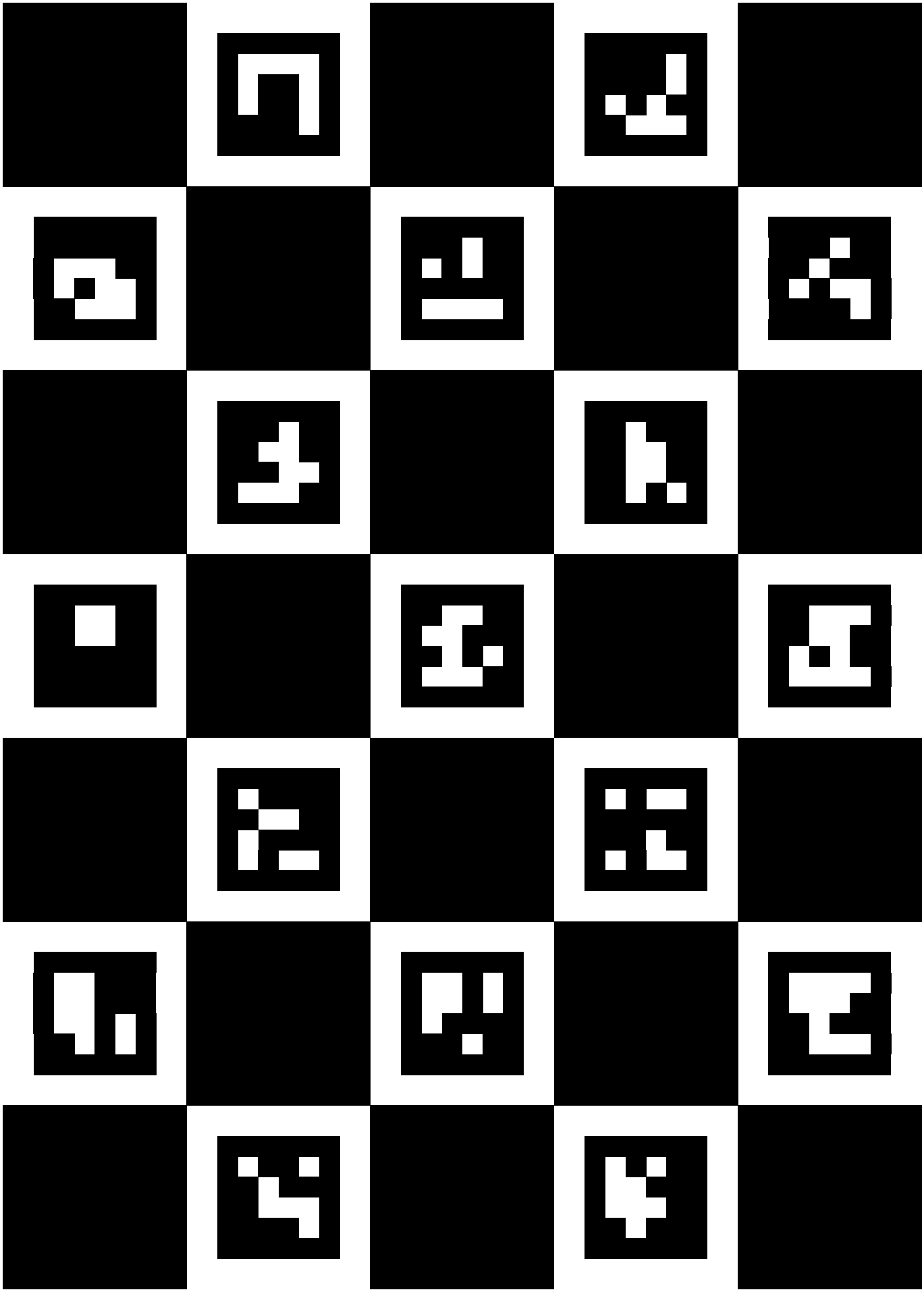
148.5 mm from either short edge

Tip is 6 mm above R for print margins

Use the paper edge, NOT the arrow tip, as R

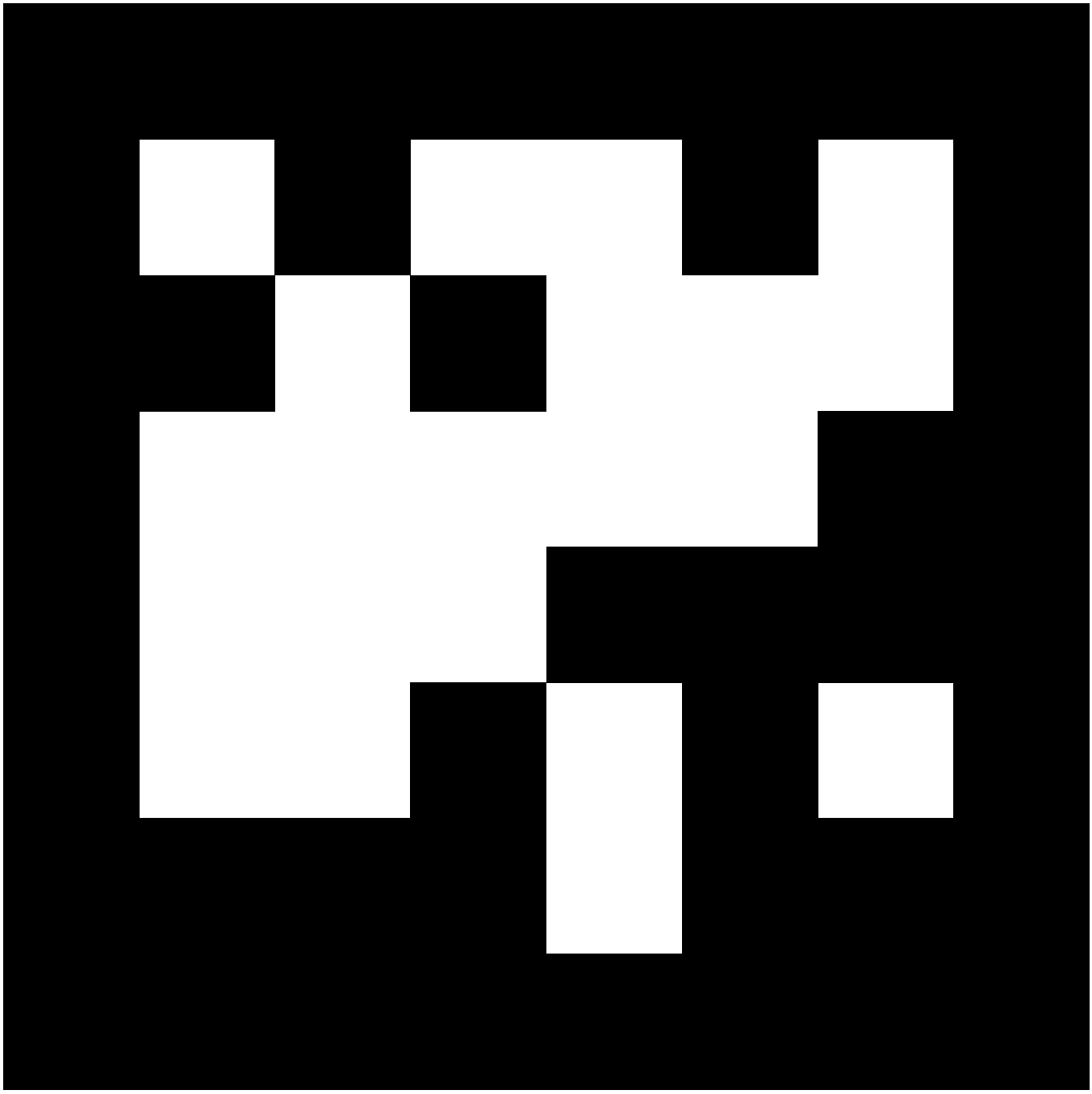
ChArUco lens / precision board

5 x 7 | square 36 mm | tag 24 mm | IDs 10-26 | modern layout



Actual size / 100%. Mount flat on rigid backing. Different size from v1.

Use close, varied views at fixed optics. Needs ChArUco detection, not ID 8 mode.



OPTIONAL APRILTAG 36h11 / ID 8

180 x 180 mm black square; keep the entire white margin clear.

R = physical bottom-left paper corner. Do not trim.

Code centre from R: (+105, +197) mm. Flat floor test only.

Compare under the same camera settings; use INSTEAD of page 12.

Requires AprilTag 36h11 detection; not enabled in the default tracker.

100 mm horizontal print check

R = PAPER CORNER

Installation and validation record

Record physical reference positions; do not fill blanks with nominal targets.

Origin: proposed near-left inside-wall reference. X right, Y far. Laser: Mileseey S6 100 m; working total allowance 1-2 mm including hand placement. Centre-based outer 3 mm offset: applied / changed / not applied. Write camera mode, zoom/focus and marker height below.

ID	R target X / Y mm	Actual R X / Y mm	Height / note
0	0 / 0	_____	_____
1	3000 / 0	_____	_____
2	3000 / 3000	_____	_____
3	0 / 3000	_____	_____
4	1000 / 1000	_____	_____
5	2000 / 1000	_____	_____
6	2000 / 2000	_____	_____
7	1000 / 2000	_____	_____
27	1500 / 1000	_____	_____
28	1500 / 2000	_____	_____

Camera A / B resolution, focus and zoom:

Physical origin and centre datum:

Inside wall span and relevant bow:

Marker / board horizontal and vertical print checks:

Independent ID 8 checks (R measured X / Y and tracked X / Y):